



Spatiotemporal evolution and driving factors of China's carbon footprint pressure: Based on vegetation carbon sequestration and LMDI decomposition

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ARTICLE INFO

Handling Editor: Isabel Soares

Keywords:

Carbon footprint pressure
Time trend
Spatial pattern
Influencing factors
Decoupling state

ABSTRACT

Evaluating carbon footprint pressure (CFP) scientifically and quantifying the environmental impact of economic activities precisely are crucial for informing policies on carbon emission reduction and achieving carbon neutrality in China. This study revealed China's CFP from 2005 to 2021 using data sets of fossil fuel carbon emissions and vegetation carbon sequestration from remote sensing. We investigated the drivers of CFP using LMDI decomposition. Additionally, we used the Tapio model to examine the decoupling of CFP and economic growth. The findings indicate that China's CFP has increased over time and shows a positive spatial correlation. Rapid economic growth and significant population expansion are the primary drivers of the surge in CFP. Conversely, optimizing energy structure, reducing energy intensity, and enhancing ecological quality are key to alleviating the rising pressure on China's carbon footprint. Moreover, we observe a weak decoupling of China's CFP and economic growth. Notably, it is more challenging to achieve an ideal decoupling state in the western region than in the central and eastern regions.

1. Introduction

Over recent decades, the rapid expansion of the global economy has been closely linked to the extensive use of energy, particularly fossil fuels such as oil and coal [1]. This surge has led to a significant increase in greenhouse gas emissions, notably carbon dioxide, which has become a major contributor to global warming [2]. To address these challenges, numerous countries have joined various climate agreements, such as the Paris Agreement and the Carbon Neutrality Alliance Statement, which provide detailed guidelines for combating global warming and climate change. As the world's largest emitter of carbon, China explicitly committed to achieving "carbon peaking and carbon neutrality" in September 2020. This highlights China's firm determination and responsibility as a major nation in actively tackling climate change. China's approach to achieving carbon neutrality provides a practical reference for other nations. It is crucial for the Chinese government to have a comprehensive understanding of existing emission reduction pressures to achieve the ambitious goal of carbon neutrality within the specified period. This understanding should form the basis for developing sustainable emission reduction strategies that align with unique

developmental characteristics.

However, reducing carbon emissions involves more than just measuring total carbon dioxide emissions. According to the British Standards Institution, carbon neutrality is achieved when human activity-generated carbon emissions and carbon dioxide absorption balance each other out (PAS 2060). Therefore, it is crucial to include total carbon absorption in the analytical framework, creating a carbon footprint pressure indicator that reflects a comprehensive energy-ecological perspective. This approach alone allows for a scientific and objective evaluation of China's carbon neutrality efforts. Vegetation carbon sequestration plays a crucial role in managing the carbon cycle of regional ecosystems and significantly reducing atmospheric carbon dioxide content [3]. This offers a solid data foundation for quantifying China's CFP. Clearly, the Chinese government recognizes the role of ecosystem carbon sinks in mitigating climate warming and views enhancing carbon sink capabilities as a key strategic initiative now and a main strategy for future carbon neutrality [4]. Consequently, clarifying the current state and spatiotemporal evolution of China's CFP and revealing the impact of human activities on this metric is crucial. This has significant theoretical and practical value in easing China's carbon

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<https://doi.org/10.1016/j.energy.2024.133299>

Received 28 January 2024; Received in revised form 19 April 2024; Accepted 27 September 2024

Available online 27 September 2024

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emission pressures and enhancing the policy framework of emission reduction and sink enhancement in the context of carbon neutrality.

2. Literature review

Clarifying the drivers of carbon emissions and developing appropriate reduction measures are key steps toward achieving carbon neutrality. In recent years, many scholars have discussed carbon emission reduction extensively [5–7]. Methodologically, these discussions fall into two categories: factor decomposition research and econometric model research. In carbon emission factor decomposition, structural decomposition analysis (SDA) faces limitations due to its dependence on detailed input-output tables [8,9]. Conversely, exponential decomposition analysis (IDA) has broader applicability [10,11]. Notably, the LMDI decomposition method, esteemed by scholars, offers quantitative insights into the factors driving changes in carbon emissions [12–14]. In econometric models, time series, panel, and spatial econometric models are widely used to explore factors influencing carbon emission reduction [15–20]. These studies confirm the effects of economic growth, population, energy use, industrial structure, and trade openness on carbon emissions, and examine the impact of macro policies like industrialization, urbanization, and emissions trading. These analyses provide a scientific basis for refining government strategies on carbon emission reduction.

Increased fossil energy consumption, driven by economic growth and a major source of greenhouse gases like carbon dioxide, has attracted significant attention from governments at all levels [21,22]. The investigation into when or if economic growth and carbon emissions reach an inflection point has become a key topic in academic discussions [23,24]. For example, Wang et al. [25] examined the presence of the Environmental Kuznets Curve (EKC) hypothesis across 208 countries worldwide. Meanwhile, many researchers have consistently supported this hypothesis in various contexts, including developed economies, EU nations, BRICS countries, developing nations, and the OECD [26–30]. However, it is challenging to use the EKC hypothesis to study the dynamic relationship between carbon emissions and GDP over different periods [31]. From a temporal perspective, the decoupling model has become a preferred approach among scholars [32,33].

These studies primarily focus on the drivers of carbon emissions and their decoupling from GDP, providing insights from an emissions standpoint but often neglecting the critical role of carbon sequestration in achieving carbon neutrality. Consequently, researchers have aimed to incorporate vegetation carbon sequestration into indicators for developing a CFP index [34]. They have also explored how economic, population, technological, and spatial agglomeration affect CFP [31,35,36]. Additionally, there is a significant scholarly focus on studying the decoupling relationship between CFP and GDP. For example, Huang et al. [37] assert that China is transitioning from weak to strong decoupling between CFP and economic growth. Furthermore, Liang et al. [38] showed that nearly half of China's provinces met their decoupling targets between 2011 and 2015. However, Yang et al. [39] indicated potential instability between CFP and GDP decoupling relationship. Despite providing key insights from an economic-environmental-energy perspective [40], these studies face several limitations in supporting carbon neutrality decision-making. First, verifying the robustness of findings from a single administrative-level sample is challenging. Second, most research uses the traditional IPAT framework, which overlooks energy, ecology, and other critical factors. Third, the research conclusions on decoupling relationships are inconsistent, and in the face of current academic doubts about the objectivity of GDP data, current research fails to provide feasible solutions.

Therefore, this study aims to enhance existing research by addressing three crucial aspects. Firstly, leveraging the inherent objectivity and timeliness of remote sensing data, we utilized the Net Primary Productivity (NPP) and the carbon emission dataset to construct the carbon

footprint pressure (CFP) index at both the national and provincial levels, revealing the overall time trend and provincial spatial pattern. Secondly, we extended the traditional IPAT framework to quantify the contributions of energy structure, energy intensity, economic growth, population size, and ecological intensity to CFP. Lastly, we investigated the decoupling relationship between CFP and GDP in China and its provinces, and used nighttime light data as a proxy variable for GDP in the robustness test to alleviate problems that may be caused by potential GDP statistical errors.

The remainder of the study is structured as follows. Section 3 provides an introduction to the IPAT equation, Kaya identity, LMDI decomposition, and Tapio decoupling methods, along with detailing all data and sources. Section 4 analyzes the spatiotemporal evolution characteristics of China's CFP, presenting all empirical results. Section 5 succinctly summarizes the primary conclusions drawn from this study, offering pertinent policy implications and outlining avenues for future research.

3. Methodology and data

3.1. IPAT framework

American ecologists Ehrlich and Holdren [41] introduced the IPAT framework. According to York et al. [42], its primary application, serving as an environmental impact analysis framework, is to assess the influence of human activities on the natural environment. This framework helps decision-makers, researchers, and policymakers gain a deeper understanding of the underlying causes of environmental degradation. The mathematical expression of the IPAT framework is as follows:

$$I = P \times A \times T \quad (1)$$

The framework reveals the environmental impact (I) from population size and distribution (P), affluence (A), and technology level (T), and is widely used in environmental policy research, resource management, and sustainable development. However, this study defines the environmental impact as China's CFP, considering both the total carbon emissions and the carbon dioxide sequestered by terrestrial vegetation through photosynthesis. To deepen the analysis, we expand the classical IPAT framework by incorporating fossil energy consumption structure (E) and ecological intensity (V). Concurrently, technological factors are reflected in the nuanced dynamics of economic growth and energy structure. The expanded analysis framework is succinctly articulated as follows:

$$I = P \times A \times E \times V \quad (2)$$

3.2. Decomposition method

3.2.1. Kaya equation

To identify the root causes of changes in China's CFP, we applied the Kaya equation, decomposing these changes into several reasonable influencing factors, building on the previously expanded IPAT framework. Concurrently, we aim to quantify each determinant's individual contribution to the overall CFP. The Kaya equation, as applied in this study, is articulated as follows:

$$\begin{aligned} CFP &= \frac{CE}{CS} = \frac{CE}{E} \times \frac{E}{G} \times \frac{G}{P} \times \frac{P}{S} \times \frac{S}{CS} \\ &= ES \times EI \times EG \times PD \times EC \end{aligned} \quad (3)$$

This equation, CFP denotes carbon footprint pressure, CE signifies carbon dioxide emissions, and CS characterizes carbon sequestration by terrestrial vegetation. Additionally, E signifies fossil energy consumption, G denotes gross domestic product, P represents population size, and S represents the aggregate area of terrestrial green vegetation. Following equation (3), the determinants influencing CFP are categorized into the

following five components:

- (1) Energy Structure Effect (ES): Pertains to alterations in carbon dioxide emissions resulting from variations in regional unit fossil energy consumption.
- (2) Energy Intensity Effect (EI): Encompasses fluctuations in the overall quantity of fossil energy consumed per unit of regional Gross Domestic Product (GDP).
- (3) Economic Growth Effect (EG): Entails shifts in the per capita economic level within the specified region.
- (4) Population Scale Effect (PD): Signifies alterations in the total permanent population per unit ecological area in the region.
- (5) Ecological Intensity Effect (EC): Denotes changes in the area of terrestrial green vegetation necessary to absorb a unit of carbon dioxide in the specified region.

3.2.2. LMDI decomposition

The Logarithmic Mean Divisia Index (LMDI) method broadly links environmental impacts, like carbon and PM 2.5 emissions, to various factors including production activities, economic growth, energy utilization, and technological progress [18]. Distinguished by its straightforward structure, effective handling of zero values, temporal independence, and capability for zero residual decomposition, the LMDI method is a robust analytical tool [43]. The LMDI decomposition of CFP in this study can be written as:

$$\Delta CFP^t = CFP^t - CFP^b = \Delta CFP_{ES}^t + \Delta CFP_{EI}^t + \Delta CFP_{EG}^t + \Delta CFP_{PD}^t + \Delta CFP_{EC}^t \quad (4)$$

Within this equation, t and b represent the reporting period and the base period, while ΔCFP_{ES}^t , ΔCFP_{EI}^t , ΔCFP_{EG}^t , ΔCFP_{PD}^t and ΔCFP_{EC}^t respectively represent the contribution of energy structure, energy intensity, economic growth, population size and ecological intensity to CFP. The specific form is as follows:

$$\Delta CFP_{ES}^t = L(CFP^t, CFP^b) \times \ln\left(\frac{ES^t}{ES^b}\right) \quad (5)$$

$$\Delta CFP_{EI}^t = L(CFP^t, CFP^b) \times \ln\left(\frac{EI^t}{EI^b}\right) \quad (6)$$

$$\Delta CFP_{EG}^t = L(CFP^t, CFP^b) \times \ln\left(\frac{EG^t}{EG^b}\right) \quad (7)$$

$$\Delta CFP_{PD}^t = L(CFP^t, CFP^b) \times \ln\left(\frac{PD^t}{PD^b}\right) \quad (8)$$

$$\Delta CFP_{EC}^t = L(CFP^t, CFP^b) \times \ln\left(\frac{EC^t}{EC^b}\right) \quad (9)$$

In the above equation:

$$L(CFP^t, CFP^b) = \begin{cases} \frac{CFP^t - CFP^b}{\ln(CFP^t) - \ln(CFP^b)}, & \text{if } CFP^t \neq CFP^b \\ CFP^t = CFP^b, & \text{if } CFP^t = CFP^b \end{cases} \quad (10)$$

The intricacies of the derivation process are expounded below, with specific reference to Eq. (A1) as presented in Appendix A.

3.2.3. Decoupling method

The decoupling theory, initially introduced by the Organization for Economic Cooperation and Development (OECD), is defined as the deliberate separation of the close link between environmental pollution and economic goods. Although the OECD decoupling index helps determine a region's decoupling status, it cannot distinctly differentiate between various decoupling types. To address these limitations, the Tapio decoupling index has been developed as an effective quantitative

tool [31]. This index enables detailed analysis of multiple decoupling states and critical value distributions (Fig. 1), with each state assigned an economic significance (see Table A1 in Appendix A).

The Tapio decoupling index, based on elasticity analysis in economics, overcomes the limitations associated with temporal factors. It effectively integrates both relative and absolute quantities, addressing the shortcomings of the OECD decoupling index, known for its sensitivity to base period selection and measurement volatility. Consequently, this study chooses the Tapio index to examine the decoupling dynamics between CFP and GDP. Furthermore, to address scholars' concerns about the objectivity of GDP data, we use average nighttime light brightness (ANLB) as a proxy for GDP to evaluate the robustness of our conclusions [44]. The precise formulation of the equation is detailed below:

$$D_t = \frac{\Delta CFP_t}{\Delta GDP_t} = \frac{(CFP_t - CFP_{t-1})/CFP_{t-1}}{(GDP_t - GDP_{t-1})/GDP_{t-1}} = \frac{c_t}{g_t} \quad (11)$$

$$D_t = \frac{\Delta CFP_t}{\Delta ANLB_t} = \frac{(CFP_t - CFP_{t-1})/CFP_{t-1}}{(ANLB_t - ANLB_{t-1})/ANLB_{t-1}} = \frac{c_t}{a_t} \quad (12)$$

In Eqs. (11) and (12), D_t is the decoupling index, c_t , g_t and a_t represent the CFP, GDP and ANLB growth rate, and t , $t-1$ represent the reporting period and the base period respectively.

3.3. Data

3.3.1. CFP

To comprehensively assess China's efforts to reduce carbon dioxide emissions, this study uses the ratio of total carbon emissions to total carbon sequestration, detailed in Eq. (13). The calculation of total carbon emissions, detailed in Eq. (14), involves multiplying the consumption of eight fossil energy sources by their respective carbon emission factors and summing the results. Simultaneously, total carbon sequestration is computed using NPP data from green vegetation. NPP, which represents the total organic dry matter produced by green plants over a specific time and area, is a key metric of vegetation's ability to sequester carbon dioxide from the atmosphere. This function plays a critical role in ecological regulation, mitigating the greenhouse effect, and maintaining the carbon cycle balance [45,46]. Guided by the chemical Eq. (15) of photosynthesis, it shows that the production of 1g of organic dry matter by green vegetation can fix approximately 1.63g of carbon dioxide in the

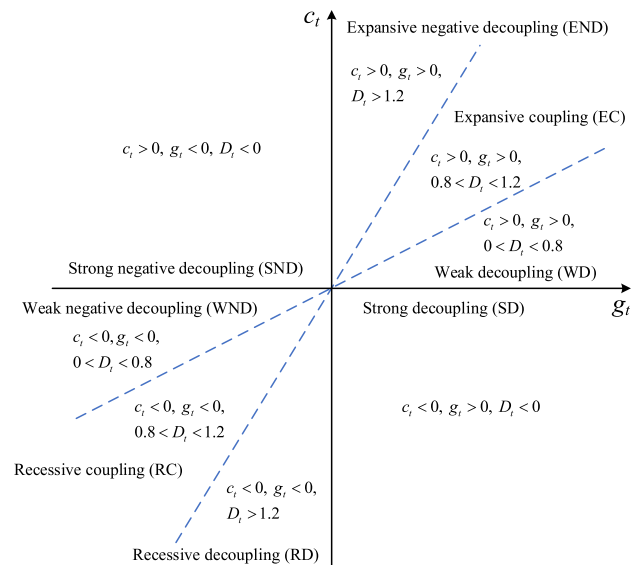


Fig. 1. Decoupled state classification diagram.

atmosphere. Using these datasets, we calculate the total carbon sequestration by combining the total area of green vegetation as detailed in Eq. (16).

$$CFP = \frac{CE}{CS} \quad (13)$$

$$CE = \sum_j \alpha_j e_j \quad (14)$$



$$CS = NPP \times S \times 1.63 \quad (16)$$

3.3.2. Data sources

The data for this study, covering fossil energy consumption, GDP, and population size from 2006 to 2022, is sourced from the China Statistical Yearbook and the China Energy Statistical Yearbook. Carbon emission factors for various energy sources are derived from the IPCC report. NPP data is sourced from NASA's MODIS product MOD17A3,¹ and the data of the designated area is extracted using ArcGIS software. Concurrently, the cumulative area of green vegetation is calculated by aggregating vector raster in the NPP data. Furthermore, we obtained continuous ANLB data by calibrating remote sensing images from DMSP-OLS and SNNP-VIIRS3.² Notably, due to a significant lack of statistical data, this study excludes Tibet, Hong Kong, Macao, and Taiwan from the research sample.

4. Empirical study

4.1. Temporal evolution trend of CFP

China's CFP shows subtle fluctuations but maintains an overall upward trajectory, as illustrated in Fig. 2. The index increased from 1.33 in 2005 to 2.32 in 2021, an approximate 75 % rise. The most rapid growth occurred between 2009 and 2010, with the index rising from 1.70 to 1.92, an annual increase of 13 %. However, the growth rate of CFP has gradually decelerated since 2012, particularly in 2013 and 2020, with noticeable intermittent declines. We explore the reasons behind this trend from both carbon emission and carbon sequestration perspectives. On one hand, Chinese government policies, including low-carbon city initiatives, new energy projects, industrial optimization, energy effi-

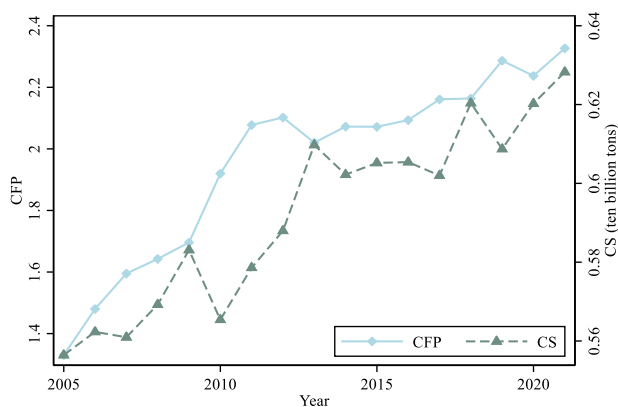


Fig. 2. Time trend of CS and CFP.

ciency enhancements, and green technology advancements, have significantly reduced carbon emissions [47–51]. On the other hand, policies enhancing ecological quality and promoting ecological civilization have also increased carbon sequestration capacity, with significant effects [52,53]. Consequently, decreases in carbon emissions or increases in carbon sequestration have contributed to the slower growth of CFP.

Despite a 2.2 % decrease in 2020, the downward trend in CFP was not sustained, as it rebounded in 2021, exceeding 2019 levels. This rebound is attributed to factors like the economic impact of the COVID-19 pandemic and a surge in production during the recovery period. Over the past decade, China's total carbon sequestration has increased by nearly 12.9 %, reaching around 720 million tons. However, the growth rate of total carbon sequestration significantly lags behind that of total carbon dioxide emissions (approximately 97 %³). This insight highlights the need for a dual focus on both carbon sources and sinks in China's pursuit of carbon neutrality. Advocating for sustainable economic development and clean energy systems must include enhancing the ecosystems' carbon sequestration capacity as a crucial goal.

Moreover, we use the Gaussian kernel density function to analyze the temporal evolution of CFP distribution characteristics at China's provincial level. Compared to the uniform kernel density function, the Gaussian kernel density function provides better smoothness and accuracy, adapting more effectively to local data nuances. As a result, the statistical outcomes more closely reflect the true probability density distribution. The density distribution of CFP across 30 provinces in 2005 and 2021 is shown in Fig. 3.

First, the core density curve from 2005 to 2021 shows a notable rightward shift, indicating a gradual increase in China's provincial CFP over the past decade. Second, provincial CFP is mainly distributed within the range of 0–10. Additionally, the 2021 peak value is lower than in 2005, indicating fewer provinces with low CFP and an increase in provinces with CFP near 10. Lastly, both curve sets show a multi-peak state, highlighting significant CFP variations across provinces.

4.2. Spatial distribution characteristics of CFP

Using the provincial averages from 2005 to 2021, we illustrate the spatial distribution of CE, CS, and CFP across China's 30 provinces in Fig. 4. Subfigure (a) illustrates the provincial-level spatial distribution of total carbon emissions (CE), revealing higher emissions in eastern provinces compared to western ones. Subfigure (b) presents the spatial

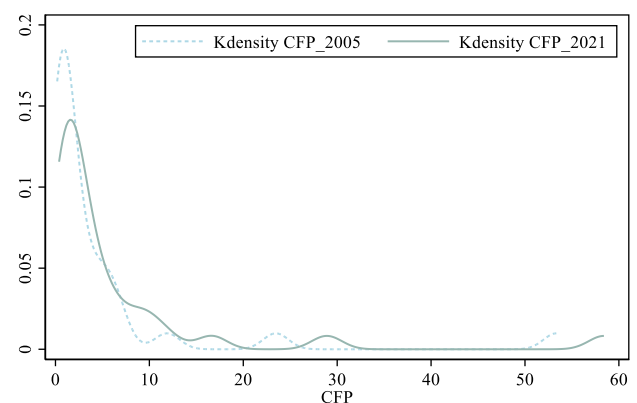


Fig. 3. Gaussian kernel density distribution of CFP.

¹ <https://doi.org/10.5067/MODIS/MOD17A3HGF.061>.

² <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/GIYGJU>.

³ The growth rate of carbon emissions from 2005 to 2021 calculated based on eight fossil energy sources.

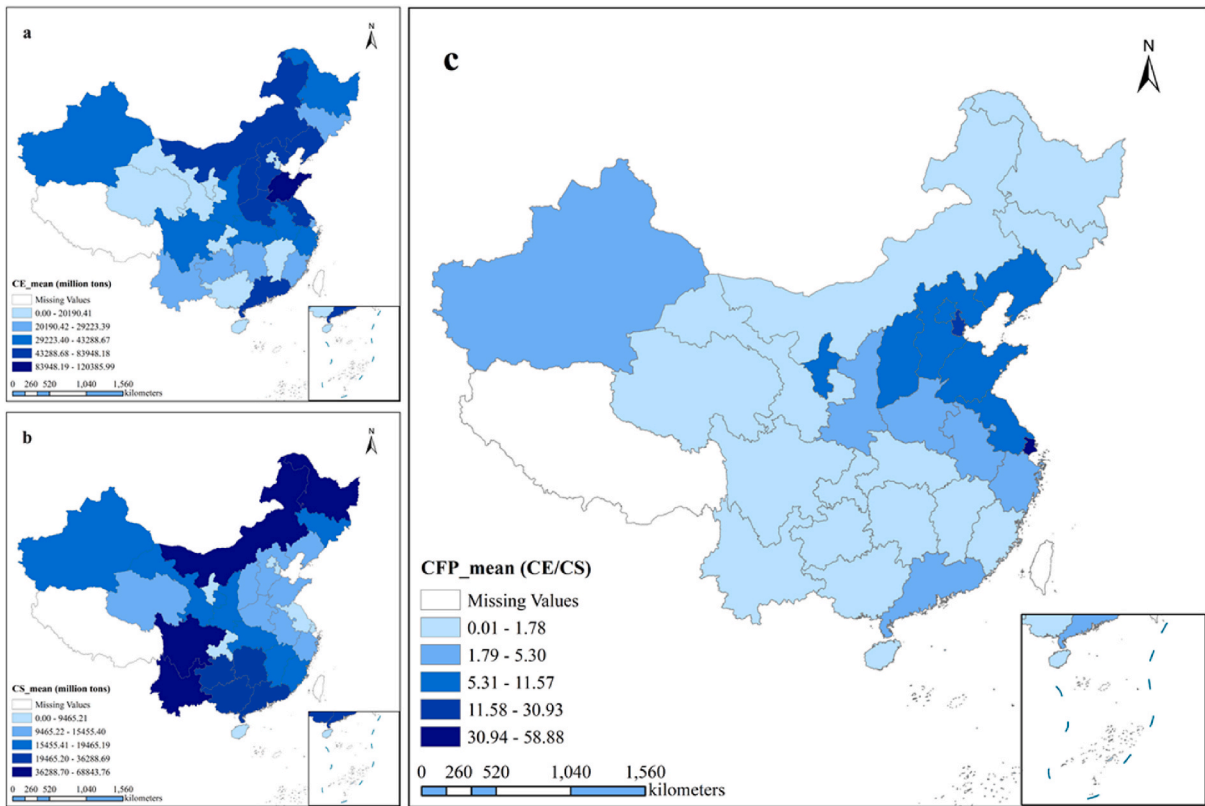


Fig. 4. Spatial distribution of China's provincial CE, CS and CFP.

distribution of total carbon sequestration (CS) at the provincial level, which is opposite to the emission patterns, being higher in central and western provinces and lower in eastern ones. Furthermore, as we expected, subfigure (c) displays a clear westward decline in CFP across the provinces of China. Specifically, 50 % of the provinces, mainly in the central and western regions, have a CFP below 2, whereas eastern coastal provinces have relatively higher CFP values. This pattern is attributed to the rapid economic development and significant increase in energy consumption in coastal cities like Shanghai, Jiangsu, Zhejiang, and Guangdong over the past decade [54]. Additionally, a significant portion of energy consumption in these regions comes from thermal power generation [55], resulting in a sharp increase in carbon dioxide emissions and a greater carbon footprint.

Furthermore, our analysis shows that the CFP of some central provinces, like Anhui and Henan, is close to that of their eastern counterparts. These populous provinces have made significant investments in industry, electricity, and transportation to meet diverse needs, leading to higher CFP [56,57]. Notably, Ningxia, a western province, has a CFP similar to eastern provinces. This is due to environmental disparities, as seen in comparisons between Ningxia and Shandong. Despite Ningxia's carbon emissions being only 16 % of Shandong's, its harsh climate and terrain limit its carbon sequestration to just 18 % of Shandong's. Finally, amidst rapid economic development and population pressure over the past decade, Shanghai faces unique challenges. Its limited land area means it has a restricted capacity for carbon sequestration, making it the province with the highest CFP in China. Its CFP is significantly higher than that of other provinces.

This study examines the detailed spatial agglomeration characteristics of China's provincial CFP using the spatial autocorrelation method within exploratory spatial data analysis [48]. Using the spatial adjacency matrix (Appendix Eq. (B3)), we measure the spatial autocorrelation of China's provincial CFP with Moran's I and Geary's C methods (Appendix Eq. (B1), Eq. (B2)). The computational results and their analysis are presented in Table 1. Notably, Moran's I consistently exceed

Table 1
Spatial autocorrelation test results of provincial CFP.

Year	Moran's I	P-value	Geary's C	P-value
2005	0.122	0.045	0.623	0.077
2006	0.139	0.035	0.610	0.061
2007	0.207	0.011	0.559	0.021
2008	0.116	0.047	0.628	0.084
2009	0.140	0.033	0.610	0.062
2010	0.134	0.047	0.617	0.062
2011	0.130	0.056	0.623	0.064
2012	0.132	0.053	0.622	0.063
2013	0.117	0.075	0.635	0.075
2014	0.152	0.055	0.617	0.042
2015	0.132	0.062	0.627	0.061
2016	0.123	0.051	0.626	0.075
2017	0.119	0.066	0.633	0.076
2018	0.127	0.063	0.629	0.067
2019	0.136	0.066	0.626	0.055
2020	0.097	0.129	0.655	0.089
2021	0.101	0.122	0.652	0.085

0, and Geary's C values fall within the [0,1] range, showing significance in most years at the 10 % level. This suggests that the spatial distribution of provincial CFP in China is not completely random but shows a spatial aggregation of similar values, indicating a positive spatial correlation [58]. Comparative analysis across different years shows subtle fluctuations in the spatial autocorrelation of China's provincial CFP.

Furthermore, Fig. 5 shows the Moran scatter plot depicting China's provincial CFP for two key years (2005 and 2021). In the plot, the horizontal axis represents the standardized provincial CFP, and the vertical axis represents the spatial lag of the provincial CFP. The results show that most provinces are located in the first and third quadrants, indicative of spatial agglomeration of similar values. The fitted line crosses the first and third quadrants, confirming the overall positive correlation in the spatial distribution of provincial CFP. Consequently,

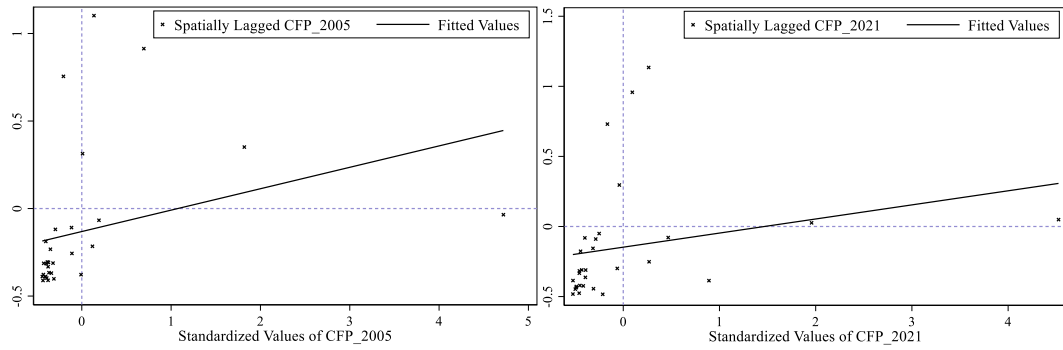


Fig. 5. Moran scatter plot of provincial CFP.

we argue that policy departments should carefully consider the spatial interactions among different provinces. This consideration could promote coordinated efforts in reducing regional carbon emissions and improving ecological quality, thereby mitigating the high CFP.

4.3. Decomposition analysis of China's CFP

Using the extended IPAT framework and previous studies [5,59–61], we performed an LMDI decomposition to examine changes in China's CFP. This comprehensive analysis covers five key dimensions: energy structure, energy intensity, economic growth, population size, and ecological intensity. Fig. 6 displays the annual decomposition results from 2005 to 2021.

Firstly, emissions from burning fossil fuels significantly impact the carbon footprint [22]. Therefore, adjusting the energy structure is crucial for reducing fossil energy consumption and environmental pollution [62]. Our findings indicate that the energy structure has consistently contributed to reducing China's CFP, reflecting a gradual improvement in energy composition over the past decade. However, it is important to note that the impact of the energy structure on CFP is relatively modest. This outcome may partly be due to the measurement methodology employed in this study. On one hand, we assert that CO₂ emissions per unit of energy are a reasonable proxy for assessing changes in the energy structure. On the other hand, limited by data availability, this study only considers the consumption of eight fossil energy sources. Therefore, the contribution of the energy structure effect might have been underestimated due to this limited scope.

Secondly, energy intensity indicates shifts in energy-saving technology and efficiency. Technological and efficiency improvements over time invariably decrease energy intensity, thus alleviating environmental pressure [63]. Our findings consistently show a negative energy intensity effect, signifying its role in curbing the rise of China's CFP. Overall, the cumulative contribution of energy intensity to China's CFP from 2005 to 2021 is 36.35 %, highlighting its significant role in mitigation.

Thirdly, research identifying sharp rises in carbon dioxide emissions and environmental pressure points to economic growth as the core factor [64]. Building on this premise, our analysis explores how economic growth contributes to shifts in China's CFP. The outcomes consistently show positive annual contributions from economic growth, underscoring that rapid development has accelerated the increase in China's CFP. Notably, in almost all years, the impact of economic growth surpasses that of other factors. From 2005 to 2021, economic growth contributed 55.6 % to CFP, nearly 20 percentage points higher than energy intensity's contribution. Therefore, in line with recent findings [65], we assert that rapid economic expansion in China over the past decade is the primary driver behind the CFP surge.

Fourthly, studies on energy consumption, carbon dioxide emissions, and environmental pollution often use the urban population ratio as a key metric to assess population size impact [66]. However, to emphasize the important role of carbon sequestration, we use the ratio of total population to terrestrial green vegetation area as a crucial indicator for assessing population size's impact on CFP. Over the years, the consistently positive population size effect indicates that China's population

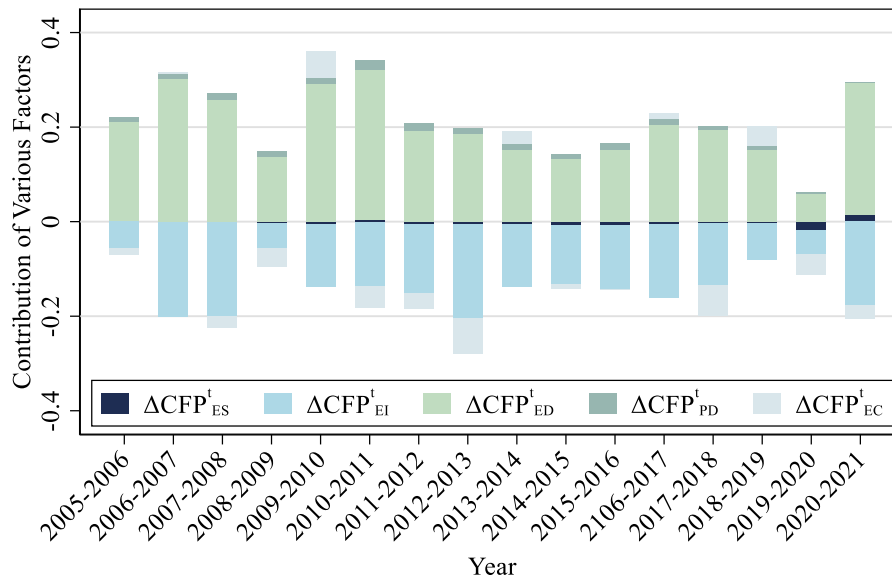


Fig. 6. LMDI decomposition results of China's CFP.

growth contributes to the increase in CFP. However, compared to energy intensity and economic growth, the population effect is modest, contributing only about 3 % to CFP.

Fifthly, as principal carbon pools, terrestrial ecosystems can sequester about 30 % of anthropogenic CO₂ emissions [67]. Higher ecological quality means less green vegetation is needed to sequester a unit of CO₂. Consequently, our study uses the ratio of total green vegetation area to the amount of CO₂ sequestered as a metric to assess the ecological intensity effect on China's CFP shifts. Notably, the ecological intensity effect is negative in most years, indicating that advancements in ecological quality significantly inhibit CFP growth. Specifically, from 2005 to 2021, China's CFP increased by about 1.00, while the vegetation area needed to fix one unit of CO₂ decreased by about 11.4 %, resulting in a CFP reduction of 0.24 over more than ten years. This underscores the crucial role of ecological protection in reducing CFP and confirms China's commitment to the concept of "harmony between man and nature".

To clarify the changes in provincial CFP in China and factor contributions, we selected 2005 as the base period and 2021 as the reporting period. We applied the LMDI method to quantify the impacts of energy structure, energy intensity, economic growth, population size, and ecological intensity on provincial CFP (Table 2). The findings show that CFP increased in 28 provinces (93.33 %) to varying degrees over the past decade. Notably, Ningxia had the highest CFP growth, increasing by approximately 11.99, while Beijing and Henan (6.67 %) saw declines in their CFP. Specifically, Beijing's CFP fell from 11.89 in 2005 to 7.48 in 2021, a significant reduction of about 37.09 %. Overall, the average CFP change across all Chinese provinces is 1.67. The added CFP of 10 provinces (33.33 %) exceeds the average, highlighting significant challenges in achieving carbon reduction and neutrality. Furthermore, in 23 provinces (76.67 %), a positive energy structure contribution has mitigated CFP, indicating that optimizing energy structure has helped control its increase. Conversely, in 7 provinces (23.33 %), including Inner Mongolia and Heilongjiang, the energy structure effect supported CFP growth. Across all provinces, energy intensity consistently exerted a

negative influence on CFP from 2005 to 2021, impeding its escalation.

Unlike energy intensity, economic growth has positively impacted all provinces' CFP, rapidly driving its increase. Shanghai shows the most significant economic growth effect, whereas Qinghai has the least impact. Regarding the population scale, 25 provinces (83.33 %) experienced a positive effect, significantly boosting their CFP. Only five provinces (16.67 %), Inner Mongolia, Liaoning, Jilin, Heilongjiang, and Gansu, experienced a negative population effect, likely due to significant outmigration [68]. Excluding Xinjiang, ecological intensity in other provinces negatively affects CFP, suggesting that ecological improvements can slow the rise of provincial CFP. The decline in ecological intensity in Xinjiang, due to vegetation degradation and rapid desertification, may significantly contribute to its CFP increase [69].

In summary, our analysis shows that economic growth and population size have accelerated CFP nationally and provincially, while energy structure, energy intensity, and ecological intensity have mitigated China's CFP. Our findings are largely consistent with related studies, specifically those of Chen et al. [34] and Peng et al. [70]. While our study provides valuable insights into China's CFP, future research should consider several key factors. For example, considering the regional structural effect could provide essential insights into the drivers of energy consumption and carbon emissions [18]. Methodologically, combining CFP research with life cycle assessment methods will likely become a prominent topic [21].

Additionally, addressing the issue of carbon inequality resulting from unequal income distribution at the micro-level could enhance the depth of future research. Factors such as the impact of investment and consumption habits among different income groups on CFP should not be overlooked [71]. In the realm of demographic factors, considering aspects like population structure, mobility, and migration could also be crucial in understanding changes in CFP [72]. Lastly, it becomes crucial to pay attention to the differentiated characteristics of different natural resources and different industries. Future research can explore ways to mitigate CFP through industrial low-carbon transformation, technological process optimization, and new energy technology development [6, 40].

Table 2
LMDI decomposition results of provincial CFP.

Province	ΔCFP^E	ΔCFP^I_{ES}	ΔCFP^I_{EI}	ΔCFP^I_{ED}	ΔCFP^I_{PD}	ΔCFP^I_{EC}
Beijing	-4.408	-2.781	-19.931	16.566	4.783	-3.045
Tianjin	5.428	-2.724	-33.473	40.428	8.240	-7.043
Hebei	1.109	-0.167	-6.832	9.165	0.534	-1.591
Shanxi	3.525	-0.077	-6.880	12.068	0.190	-1.776
Inner Mongolia	1.228	0.005	-0.740	2.227	-0.016	-0.248
Liaoning	2.086	-0.112	-3.747	6.764	-0.042	-0.778
Jilin	0.270	-0.001	-1.538	2.045	-0.169	-0.067
Heilongjiang	0.247	0.015	-0.528	0.964	-0.149	-0.055
Shanghai	5.070	-3.005	-79.714	75.265	15.881	-3.357
Jiangsu	3.298	-0.366	-10.208	14.273	0.966	-1.367
Zhejiang	1.890	-0.169	-3.647	5.139	0.960	-0.393
Anhui	1.426	-0.077	-3.017	5.097	0.010	-0.587
Fujian	1.142	-0.062	-1.391	2.466	0.229	-0.100
Jiangxi	0.503	-0.025	-1.268	1.872	0.039	-0.115
Shandong	5.059	-0.606	-8.700	15.440	0.996	-2.072
Henan	-0.030	-0.179	-6.528	7.307	0.243	-0.873
Hubei	0.458	-0.074	-2.746	3.568	0.036	-0.325
Hunan	0.264	-0.058	-1.898	2.298	0.054	-0.131
Guangdong	1.325	-0.060	-2.140	2.987	0.684	-0.145
Guangxi	0.485	-0.016	-0.408	0.929	0.047	-0.068
Hainan	1.100	0.007	-0.989	1.939	0.248	-0.105
Chongqing	0.398	-0.140	-2.553	3.090	0.223	-0.221
Sichuan	0.150	-0.069	-1.079	1.374	0.014	-0.089
Guizhou	0.438	-0.052	-1.715	2.245	0.052	-0.092
Yunnan	0.124	-0.019	-0.524	0.662	0.016	-0.011
Shaanxi	1.669	0.053	-2.046	3.875	0.152	-0.366
Gansu	0.447	0.008	-1.180	1.853	-0.027	-0.207
Qinghai	0.211	-0.014	-0.337	0.570	0.028	-0.036
Ningxia	11.991	0.158	-3.738	14.953	1.640	-1.022
Xinjiang	3.103	0.164	-0.779	3.124	0.521	0.073

4.4. The decoupling effect of CFP and GDP

According to our findings and related research [31,37], economic growth is the primary driver of the increase in CFP. However, the relationship between economic growth and CFP across different periods and administrative levels has been little studied. To bridge this gap, we use the Tapio decoupling index to analyze the decoupling dynamics between CFP and GDP in China and its 30 provinces from 2005 to 2021. Detailed results are provided in Table 3.

The findings indicate a strong decoupling of CFP from GDP during

Table 3
The decoupling state between China's CFP and GDP.

Year	C_t	g_t	C_t/g_t	Decoupling state
2005–2006	0.113	0.169	0.669	WD
2006–2007	0.078	0.225	0.347	WD
2007–2008	0.030	0.183	0.164	WD
2008–2009	0.032	0.093	0.344	WD
2009–2010	0.132	0.185	0.714	WD
2010–2011	0.082	0.185	0.445	WD
2011–2012	0.012	0.105	0.114	WD
2012–2013	-0.039	0.101	-0.386	SD
2013–2014	0.026	0.084	0.310	WD
2014–2015	-0.0003	0.070	-0.004	SD
2015–2016	0.010	0.083	0.120	WD
2016–2017	0.032	0.108	0.296	WD
2017–2018	0.001	0.098	0.010	WD
2018–2019	0.057	0.075	0.760	WD
2019–2020	-0.022	0.028	-0.786	SD
2020–2021	0.040	0.131	0.305	WD

three periods (18.75 %): 2012–2013, 2014–2015, and 2019–2020. This signifies that in these intervals, China achieved GDP growth while CFP declined. In the remaining 81.25 % of the period, a weak decoupling state occurred, where both GDP and CFP grew simultaneously, though CFP’s growth rate was slightly lower. Overall, China’s CFP and GDP predominantly exhibit a weak decoupling state. This finding aligns with the research by Chen et al. [34]. Additionally, this is linked to policy tools like industrial transformation, ecological protection, energy efficiency improvement, and clean energy development that China has implemented over the past decade [48,52].

Moreover, the decoupling results after using ANLB as a proxy variable for economic growth are shown in Table 4. We found that in 62.5 % of years, CFP and economic growth showed weak or strong decoupling. At the same time, the two years that showed strong decoupling are also consistent with the results using GDP data, which supports the results in Table 3 to a certain extent. It is worth noting that 2010 became a key node in the decoupling state of China’s CFP and economic growth. Before this year, there was great instability in the decoupling state of CFP and economic growth, which is similar to the view of Yang et al. [39]. However, after 2010, the relationship between CFP and economic growth exhibited greater stability, with most years oscillating between weak and strong decoupling. This further reinforces the notion that China is navigating a transitional phase between weak and strong decoupling of CFP and economic growth [37].

In order to carefully examine the time evolution trends and heterogeneous characteristics of the decoupling state of CFP and GDP in different provinces in China, we chose 2005–2006 and 2020–2021 as example periods to create a map depicting the decoupling dynamics between CFP and GDP (Fig. 7). The results show that during 2005–2006, all 30 provinces in China were in an expansive coupling state between CFP and economic growth. While each province’s economy grew, CFP also increased, with similar rates of change for both variables. From 2020 to 2021, 10 provinces (33.33 %) achieved strong decoupling, with GDP increasing and CFP decreasing. The other 66.67 % of provinces remained in an expansive coupling state. Over time, the shift from an extensive to an intensive economic model is expected to transition the relationship between CFP and GDP from expansive to strong decoupling [73]. Consequently, the relationship between provincial CFP and economic growth in China generally follows the expected evolutionary path.

From a regional heterogeneity perspective, 30 % of the eastern provinces have achieved strong decoupling between CFP and GDP. About 67 % of provinces in the Northeast and Central regions have achieved strong decoupling. In contrast, only about 8 % of Western provinces have reached this decoupling status. The likely explanation is the more advanced economic development in the central and eastern regions, which balances energy-saving technologies and ecological

protection with steady economic growth. Most Western provinces are still urbanizing and attract fewer mobile factors like capital, labor, and technology. Challenging natural conditions, like sparse vegetation and limited carbon sequestration capacity, make strong decoupling between CFP and GDP more difficult in the Western region.

5. Conclusion and policy implications

5.1. Conclusion and discussion

This study integrates data on carbon emissions and carbon sequestration to construct CFP datasets at both the national and provincial levels, revealing their spatiotemporal characteristics. Additionally, we employed the LMDI method to conduct a multidimensional decomposition of CFP changes and elucidate the various factors affecting it. Furthermore, in conjunction with the results of the LMDI decomposition and scholars’ focus on decoupling research [38], we analyzed the decoupling status of GDP and CFP at both the national and provincial levels. The main findings are as follows:

Firstly, China’s overall CFP has shown a steady increase over the years, primarily due to the imbalanced growth rates of carbon emissions and sequestration. This trend is driven by rapid industrialization, urbanization, higher energy consumption, and substantial reliance on fossil fuels [74,75]. Secondly, provincially, CFP tends to decrease from east to west, a trend likely due to higher economic development and energy consumption in the east. Additionally, the limited ecological capacity in the east poses challenges in curbing the rise of CFP [31]. Spatially, provincial CFP shows positive correlation clustering, suggesting that neighboring provinces face similar pressures, likely due to comparable industrial structures and resource utilization patterns [76]. Lastly, LMDI decomposition identifies rapid GDP and population growth as the primary drivers of increasing CFP, corroborated by other studies [37]. Optimizing energy structures, reducing energy intensity, and enhancing ecological quality emerge as crucial strategies to mitigate CFP, reflecting the Chinese government’s efforts in energy transition and ecological civilization development [53,77]. Furthermore, most years show a weak decoupling between China’s CFP and GDP, a finding also reported by Chen et al. [34]. Simultaneously, western provinces face greater challenges in achieving decoupling compared to their eastern counterparts.

5.2. Policy implications

The research findings in this article are practically significant for addressing CFP challenges and advancing carbon neutrality. Firstly, our research shows that China’s steadily increasing CFP over the past decade poses a significant barrier to achieving carbon neutrality. Therefore, from a temporal perspective, it is paramount to address the carbon imbalance caused by divergent growth rates in emissions and sequestration. Concurrently, policymakers should acknowledge significant provincial disparities and decentralize policy objectives to the provincial level, adhering to the principle of “tailoring measures to local conditions”. Specifically, provinces with higher CFP should implement more stringent emission reduction regulations while ensuring economic stability, whereas provinces with lower CFP should prioritize sustaining their trajectory of sustainable economic development.

Secondly, the spatial correlation in provincial CFP suggests that mitigating CFP requires collaborative regional efforts rather than isolated actions, emphasizing the importance of maximizing regional interests. Hence, to mitigate the negative externalities associated with CFP, governmental agencies must establish mechanisms for coordinating interests in regional carbon dioxide emission reduction and ecological preservation. These efforts aim to minimize costs for individuals within a region while considering the environmental constraints of the entire area. Through joint planning and collaborative governance, the objective of mitigating CFP can be effectively pursued.

Table 4
The decoupling state between China’s CFP and ANLB.

Year	c_t	a_t	c_t/a_t	Decoupling state
2005–2006	0.113	0.099	1.141	EC
2006–2007	0.078	0.029	2.690	END
2007–2008	0.030	0.030	1.000	EC
2008–2009	0.032	−0.095	−0.337	SND
2009–2010	0.132	0.274	0.482	WD
2010–2011	0.082	0.148	0.554	WD
2011–2012	0.012	0.021	0.571	WD
2012–2013	−0.039	0.088	−0.443	SD
2013–2014	0.026	0.072	0.361	WD
2014–2015	−0.0003	0.013	−0.023	SD
2015–2016	0.010	−0.011	−0.909	SND
2016–2017	0.032	0.259	0.124	WD
2017–2018	0.001	0.004	0.250	WD
2018–2019	0.057	0.086	0.663	WD
2019–2020	−0.022	−0.045	0.489	WND
2020–2021	0.040	0.082	0.488	WD

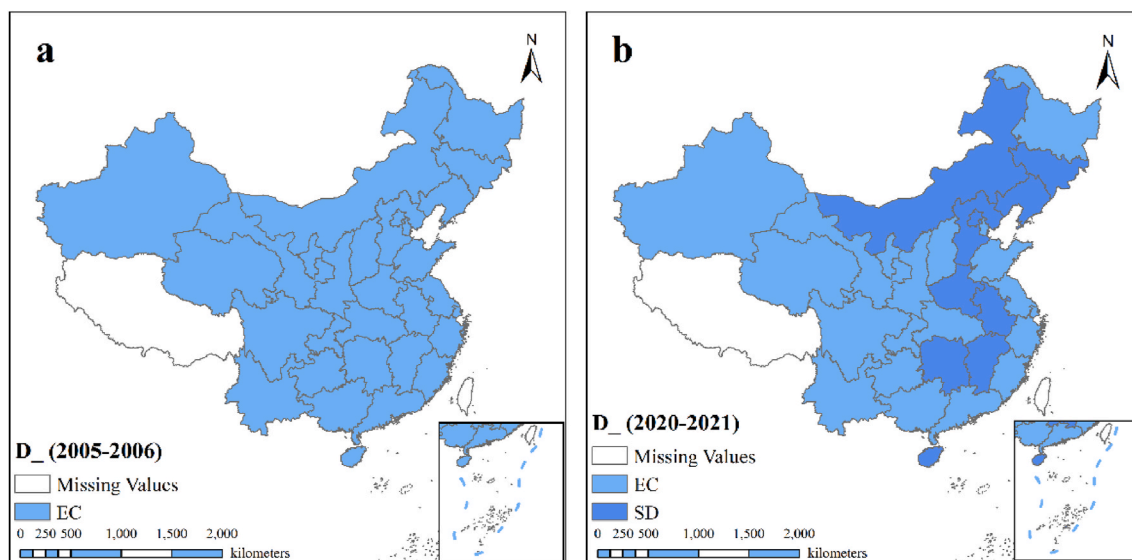


Fig. 7. Decoupled state between provincial CFP and GDP.

Thirdly, previous LMDI decomposition analyses underscore the need for a comprehensive policy framework to curb the rise in CFP. On one hand, economic growth and population size have been shown to positively impact CFP. Key measures to mitigate CFP include eliminating outdated production capacities, promoting green industry transformation, raising awareness about energy conservation and environmental preservation, and encouraging green and low-carbon travel methods. On the other hand, the significant impact of fossil fuel dependency and energy efficiency on CFP must not be underestimated. Consequently, promoting renewable energy use and supporting green technology research and development can significantly reduce CFP. Additionally, our findings highlight the considerable positive impact of green spaces on CFP throughout all periods. Therefore, it is critical that China maintains a strong carbon sequestration capability through initiatives such as expanding green spaces, protecting natural habitats, and carrying out ecological rehabilitation projects.

Finally, while China has achieved a weak decoupling between GDP and CFP nationally, western provinces continue to face decoupling challenges. Consequently, the western region must balance economic development with environmental sustainability, necessitating a fundamental shift from traditional development paradigms. Fortunately, with their abundant wind and solar energy resources, western provinces have promising opportunities to foster green economic growth. Improving the stability of renewable energy supply, its utilization efficiency, and green power consumption are critical for determining if western provinces can achieve future GDP and CFP decoupling under China's energy transition policy.

5.3. Research shortcomings and prospects

While this study advances the research landscape through innovations in index construction, factor decomposition, and decoupling analysis, some aspects require further investigation. For instance, we conducted a comprehensive analysis of the factors driving China's CFP growth using LMDI decomposition. Nevertheless, this study does not sufficiently detail the complex impact mechanisms of each element on CFP. Consequently, our future research endeavors will delve deeper into these intricacies.

CRediT authorship contribution statement

Lingyue Tian: Writing – original draft, Methodology. **Jian Chai:** Project administration, Conceptualization. **Xiaokong Zhang:** Software.

Yue Pan: Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We have indicated all data sources in the manuscript.

Acknowledgments

This work was supported by Xidian University, the National Natural Science Foundation of China under Grants No. 72373116, the Natural Science Basic Research Program of Shaanxi under Grants No. 2024JC-JCQN-40, the Youth Innovation Team of Shaanxi Universities (2020-68), and Shaanxi Province Qin Chuangyuan “scientist + engineer” team building project (2022KXJ-007).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2024.133299>.

References

- [1] Chen H, Yi J, Chen A, Peng D, Yang J. Green technology innovation and CO₂ emission in China: evidence from a spatial-temporal analysis and a nonlinear spatial durbin model. *Energy Pol* 2023;172:113338.
- [2] Krevor S, De Coninck H, Gasda SE, Ghaleigh NS, de Gooyert V, Hajibeygi H, Juanes R, Neufeld J, Roberts JJ, Swennenhuis F. Subsurface carbon dioxide and hydrogen storage for a sustainable energy future. *Nat Rev Earth Environ* 2023;4(2): 102–18.
- [3] Fu D, Bu B, Wu J, Singh RP. Investigation on the carbon sequestration capacity of vegetation along a heavy traffic load expressway. *J Environ Manag* 2019;241: 549–57.
- [4] Cheng S, Chen Y, Meng F, Chen J, Liu G, Song M. Impacts of local public expenditure on CO₂ emissions in Chinese cities: a spatial cluster decomposition analysis. *Resour Conserv Recycl* 2021;164:105217.
- [5] Xu SC, He ZX, Long RY. Factors that influence carbon emissions due to energy consumption in China: decomposition analysis using LMDI. *Appl Energy* 2014;127: 182–93.
- [6] Wu L, Sun L, Qi P, Ren X, Sun X. Energy endowment, industrial structure upgrading, and CO₂ emissions in China: revisiting resource curse in the context of carbon emissions. *Resour Pol* 2021;74:102329.

- [7] Xie P, Gong N, Sun F, Li P, Pan X. What factors contribute to the extent of decoupling economic growth and energy carbon emissions in China? *Energy Pol* 2023;173:113416.
- [8] Chen X, Rahaman MA, Murshed M, Mahmood H, Hossain MA. Causality analysis of the impacts of petroleum use, economic growth, and technological innovation on carbon emissions in Bangladesh. *Energy* 2023;267:126565.
- [9] Wang W, Hu Y, Lu Y. Driving forces of China's provincial bilateral carbon emissions and the redefinition of corresponding responsibilities. *Sci Total Environ* 2023;857:159404.
- [10] Hussain J, Pan Y, Ali G, Xiaofang Y. Pricing behavior of monopoly market with the implementation of green technology decision under emission reduction subsidy policy. *Sci Total Environ* 2020;709:136110.
- [11] Chu X, Du G, Geng H, Liu X. Can energy quota trading reduce carbon intensity in China? A study using a DEA and decomposition approach. *Sustain Prod Consum* 2021;28:1275–85.
- [12] Ang BW. Monitoring changes in economy-wide energy efficiency: from energy-GDP ratio to composite efficiency index. *Energy Pol* 2006;34(5):574–82.
- [13] Shao S, Yang L, Gan C, Cao J, Geng Y, Guan D. Using an extended LMDI model to explore techno-economic drivers of energy-related industrial CO₂ emission changes: a case study for Shanghai (China). *Renew Sustain Energy Rev* 2016;55: 516–36.
- [14] Isik M, Sarica K, Ari I. Driving forces of Turkey's transportation sector CO₂ emissions: an LMDI approach. *Transport Pol* 2020;97:210–9.
- [15] Khan Z, Ali S, Umar M, Kirikkaleli D, Jiao Z. Consumption-based carbon emissions and international trade in G7 countries: the role of environmental innovation and renewable energy. *Sci Total Environ* 2020;730:138945.
- [16] Ouyang X, Fang X, Cao Y, Sun C. Factors behind CO₂ emission reduction in Chinese heavy industries: do environmental regulations matter? *Energy Pol* 2020;145: 111765.
- [17] Abbasi KR, Shahbaz M, Jiao Z, Tufail M. How energy consumption, industrial growth, urbanization, and CO₂ emissions affect economic growth in Pakistan? A novel dynamic ARDL simulations approach. *Energy* 2021;221:119793.
- [18] Wang S, Sun S, Zhao E, Wang S. Urban and rural differences with regional assessment of household energy consumption in China. *Energy* 2021;232:121091.
- [19] Chien F, Hsu CC, Ozturk I, Sharif A, Sadiq M. The role of renewable energy and urbanization towards greenhouse gas emission in top Asian countries: evidence from advance panel estimations. *Renew Energy* 2022;186:207–16.
- [20] Shu H, Wang Y, Umar M, Zhong Y. Dynamics of renewable energy research, investment in EnvoTech and environmental quality in the context of G7 countries. *Energy Econ* 2023;120:106582.
- [21] Nowrouzi M, Abyar H. A framework for the design and optimization of integrated fixed-film activated sludge-membrane bioreactor configuration by focusing on cost-coupled life cycle assessment. *J Clean Prod* 2021;296:126557.
- [22] Abyar H, Nowrouzi M. A comprehensive framework for eco-environmental impact evaluation of wastewater treatment plants: integrating carbon footprint, energy footprint, toxicity, and economic assessments. *J Environ Manag* 2023;348:119255.
- [23] Riti JS, Song D, Shu Y, Kamah M. Decoupling CO₂ emission and economic growth in China: is there consistency in estimation results in analyzing environmental Kuznets curve? *J Clean Prod* 2017;166:1448–61.
- [24] Ridzuan NHAM, Marwan NF, Khalid N, Ali MH, Tseng ML. Effects of agriculture, renewable energy, and economic growth on carbon dioxide emissions: evidence of the environmental Kuznets curve. *Resour Conserv Recycl* 2020;160:104879.
- [25] Wang Q, Zhang F, Li R. Revisiting the environmental kuznets curve hypothesis in 208 countries: the roles of trade openness, human capital, renewable energy and natural resource rent. *Environ Res* 2023;216:114637.
- [26] Xia W, Apergis N, Bashir MF, Ghosh S, Doğan B, Shahzad U. Investigating the role of globalization, and energy consumption for environmental externalities: empirical evidence from developed and developing economies. *Renew Energy* 2022;183:219–28.
- [27] Bölük G, Mert M. Fossil & renewable energy consumption, GHGs (greenhouse gases) and economic growth: evidence from a panel of EU (European Union) countries. *Energy* 2014;74:439–46.
- [28] Chien F, Anwar A, Hsu CC, Sharif A, Razzaq A, Sinha A. The role of information and communication technology in encountering environmental degradation: proposing an SDG framework for the BRICS countries. *Technol Soc* 2021;65: 101587.
- [29] Akram R, Chen F, Khalid F, Ye Z, Majeed MT. Heterogeneous effects of energy efficiency and renewable energy on carbon emissions: evidence from developing countries. *J Clean Prod* 2020;247:119122.
- [30] Churchill SA, Inekwe J, Ivanovski K, Smyth R. The environmental Kuznets curve in the OECD: 1870–2014. *Energy Econ* 2018;75:389–99.
- [31] Fan W, Huang S, Yu Y, Xu Y, Cheng S. Decomposition and decoupling analysis of carbon footprint pressure in China's cities. *J Clean Prod* 2022;372:133792.
- [32] Zhang YJ, Da YB. The decomposition of energy-related carbon emission and its decoupling with economic growth in China. *Renew Sustain Energy Rev* 2015;41: 1255–66.
- [33] Yang J, Cai W, Ma M, Li L, Liu C, Ma X, Li L, Chen X. Driving forces of China's CO₂ emissions from energy consumption based on Kaya-LMDI methods. *Sci Total Environ* 2020;711:134569.
- [34] Chen J, Fan W, Li D, Liu X, Song M. Driving factors of global carbon footprint pressure: based on vegetation carbon sequestration. *Applied energy* 2020;267: 114914.
- [35] Liang D, Lu H, Guan Y, Feng L, Chen Y, He L. Further mitigating carbon footprint pressure in urban agglomeration by enhancing the spatial clustering. *J Environ Manag* 2023;326:116715.
- [36] Liang D, Lu H, Guan Y, Feng L, He L, Qiu L, Lu J. Population density regulation may mitigate the imbalance between anthropogenic carbon emissions and vegetation carbon sequestration. *Sustain Cities Soc* 2023;92:104502.
- [37] Huang Y, Yu Q, Wang R. Driving factors and decoupling effect of carbon footprint pressure in China: based on net primary production. *Technol Forecast Soc Change* 2021;167:120722.
- [38] Liang D, Lu H, Guan Y, Feng L. Drivers for decoupling carbon footprint pressure from economic growth in China's provinces. *Geography and Sustainability* 2022;3 (3):258–67.
- [39] Yang Y, Li Y, Guo Y. Impact of the differences in carbon footprint driving factors on carbon emission reduction of urban agglomerations given SDGs: a case study of the Guanzhong in China. *Sustain Cities Soc* 2022;85:104024.
- [40] Nowrouzi M, Abyar H, Rohani S. A comparison of nitrogen removal systems through cost-coupled life cycle assessment and energy efficiency analysis. *Sci Total Environ* 2023;858:159787.
- [41] Ehrlich PR, Holdren JP. Impact of Population Growth: complacency concerning this component of man's predicament is unjustified and counterproductive. *Science* 1971;171(3977):1212–7.
- [42] York R, Rosa EA, Dietz T. Bridging environmental science with environmental policy: plasticity of population, affluence, and technology. *Soc Sci Q* 2002;83(1): 18–34.
- [43] Ang BW. Decomposition analysis for policymaking in energy: which is the preferred method? *Energy Pol* 2004;32(9):1131–9.
- [44] Bennett MM, Smith LC. Advances in using multitemporal night-time lights satellite imagery to detect, estimate, and monitor socioeconomic dynamics. *Remote Sensing of Environment* 2017;192:176–97.
- [45] Michaletz ST, Cheng D, Kerkhoff AJ, Enquist BJ. Convergence of terrestrial plant production across global climate gradients. *Nature* 2014;512(7512):39–43.
- [46] Song C, Dodds WK, Rüegg J, Argerich A, Baker CL, Bowden WB, Douglas MM, Farrell KJ, Plinn MB, Garcia EA, Helton AM. Continental-scale decrease in net primary productivity in streams due to climate warming. *Nat Geosci* 2018;11(6): 415–20.
- [47] Huo W, Qi J, Yang T, Liu J, Liu M, Zhou Z. Effects of China's pilot low-carbon city policy on carbon emission reduction: a quasi-natural experiment based on satellite data. *Technol Forecast Soc Change* 2022;175:121422.
- [48] Chai J, Tian L, Jia R. New energy demonstration city, spatial spillover and carbon emission efficiency: evidence from China's quasi-natural experiment. *Energy Pol* 2023;173:113389.
- [49] Zhao J, Jiang Q, Dong X, Dong K, Jiang H. How does industrial structure adjustment reduce CO₂ emissions? Spatial and mediation effects analysis for China. *Energy Econ* 2022;105:105704.
- [50] Liu D. Convergence of energy carbon emission efficiency: evidence from manufacturing sub-sectors in China. *Environ Sci Pollut Control Ser* 2022;29(21): 31133–47.
- [51] Shao S, Liu J, Geng Y, Miao Z, Yang Y. Uncovering driving factors of carbon emissions from China's mining sector. *Appl Energy* 2016;166:220–38.
- [52] Rong T, Zhang P, Zhu H, Jiang L, Li Y, Liu Z. Spatial correlation evolution and prediction scenario of land use carbon emissions in China. *Ecol Inf* 2022;71: 101802.
- [53] Jiang L, Zhao W, Lewis BJ, Wei Y, Dai L. Effects of management regimes on carbon sequestration under the Natural Forest Protection Program in northeast China. *J For Res* 2018;29:1187–94.
- [54] Feng Q, Sun T, Liu X. Collaborative governance on SO₂ pollution from thermal power energy consumption: evidence from Yangtze River Delta urban agglomeration in China. *Energy Environ* 2023;0958305X231192351.
- [55] Pu Y, Song J, Dong L, Yang W, Wang S. Estimating mitigation potential and cost for air pollutants of China's thermal power generation: a GAINS-China model-based spatial analysis. *J Clean Prod* 2019;211:749–64.
- [56] Cui P, Xia S, Hao L. Do different sizes of urban population matter differently to CO₂ emission in different regions? Evidence from electricity consumption behavior of urban residents in China. *J Clean Prod* 2019;240:118207.
- [57] Zhao P, Zeng L, Li P, Lu H, Hu H, Li C, Zheng M, Li H, Yu Z, Yuan D, Xie J. China's transportation sector carbon dioxide emissions efficiency and its influencing factors based on the EBM DEA model with undesirable outputs and spatial Durbin model. *Energy* 2022;238:121934.
- [58] Zhang Y, Pan J, Zhang Y, Xu J. Spatial-temporal characteristics and decoupling effects of China's carbon footprint based on multi-source data. *J Geogr Sci* 2021; 31:327–49.
- [59] Fan Y, Liu LC, Wu G, Tsai HT, Wei YM. Changes in carbon intensity in China: empirical findings from 1980–2003. *Ecol Econ* 2007;62(3–4):683–91.
- [60] Wang C, Chen J, Zou J. Decomposition of energy-related CO₂ emission in China: 1957–2000. *Energy* 2005;30(1):73–83.
- [61] Miksa O, Chen X, Baležentienė L, Streimikiene D, Baležentis T. Ecological challenges in life cycle assessment and carbon budget of organic and conventional agroecosystems: a case from Lithuania. *Sci Total Environ* 2020;714:136850.
- [62] De Freitas LC, Kaneko S. Decomposing the decoupling of CO₂ emissions and economic growth in Brazil. *Ecol Econ* 2011;70(8):1459–69.
- [63] Huang J, Du D, Tao Q. An analysis of technological factors and energy intensity in China. *Energy Pol* 2017;109:1–9.
- [64] Li J, Li S. Energy investment, economic growth and carbon emissions in China—empirical analysis based on spatial Durbin model. *Energy Pol* 2020;140: 111425.
- [65] Meng Y, Wu H, Wang Y, Duan Y. International trade diversification, green innovation, and consumption-based carbon emissions: the role of renewable energy for sustainable development in BRICST countries. *Renew Energy* 2022;198: 1243–53.

- [66] Zheng W, Walsh PP. Economic growth, urbanization and energy consumption—a provincial level analysis of China. *Energy Econ* 2019;80:153–62.
- [67] Pan Y, Birdsey RA, Fang J, Houghton R, Kauppi PE, Kurz WA, Phillips OL, Shvidenko A, Lewis SL, Canadell JG, Ciais P. A large and persistent carbon sink in the world's forests. *Science* 2011;333(6045):988–93.
- [68] Ma Z, Li C, Zhang J. Understanding urban shrinkage from a regional perspective: case study of Northeast China. *J Urban Plann Dev* 2020;146(4):05020025.
- [69] Ma X, Zhu J, Yan W, Zhao C. Projections of desertification trends in Central Asia under global warming scenarios. *Sci Total Environ* 2021;781:146777.
- [70] Peng D, Yi J, Chen A, Chen H, Yang J. Factor decomposition for ecological pressure of the whole industrial energy carbon footprint: a case study of China. *Environ Sci Pollut Control Ser* 2023;30(12):33862–76.
- [71] Chancel L. Global carbon inequality over 1990–2019. *Nat Sustain* 2022;5(11):931–8.
- [72] Carleton TA, Hsiang SM. Social and economic impacts of climate. *Science*. 2016; 353(6304):aad9837.
- [73] Zhang YJ, Hao JF. Carbon emission quota allocation among China's industrial sectors based on the equity and efficiency principles. *Ann Oper Res* 2017;255:117–40.
- [74] Sun W, Huang C. How does urbanization affect carbon emission efficiency? Evidence from China. *J Clean Prod* 2020;272:122828.
- [75] Appiah M, Li M, Naeem MA, Karim S. Greening the globe: uncovering the impact of environmental policy, renewable energy, and innovation on ecological footprint. *Technol Forecast Soc Change* 2023;192:122561.
- [76] Cai X, Wang W, Rao A, Rahim S, Zhao X. Regional sustainable development and spatial effects from the perspective of renewable energy. *Front Environ Sci* 2022;10:166.
- [77] Li R, Wang X, Wang Q. Does renewable energy reduce ecological footprint at the expense of economic growth? An empirical analysis of 120 countries. *J Clean Prod* 2022;346:131207.